

Reflection Report: OOP Final Project

Team Members: B123245013 劉邦均, B123040009 陳予涵, B123040027 林昱辰

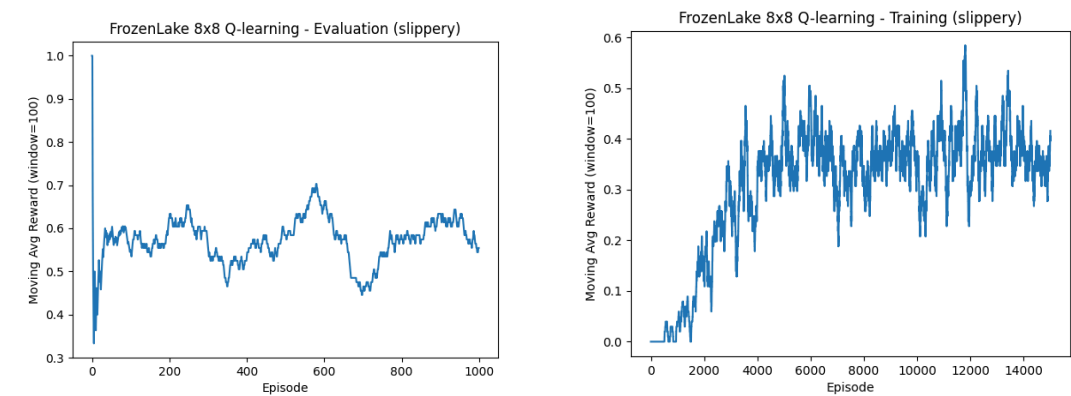
Part 2: Frozen Lake Q-Learning Optimization

Improved Version (`john_version.py`)

Key Improvements

Feature	Original	Improved	Impact
Q-table Init	0	5.0	↑ Early exploration
Learning Rate	0.9 (fixed)	0.3→0.05 (decay)	↑ Convergence stability
Epsilon Decay	Linear	Exponential	↑ Exploration efficiency
Discount Factor	0.9	0.99	↑ Long-term planning
Reward Shaping	None	Shaped	↑ Learning speed
Evaluation	Basic	ϵ -controllable	↑ Evaluation accuracy

Training Results



Training Phase Observations:

- **First 5,000 Episodes:** Success rate gradually increases from 0%
- **5,000-10,000 Episodes:** Accelerated improvement, moving average reaches 0.4-0.6
- **10,000-15,000 Episodes:** Converges to stable level

Evaluation Results (1000 test episodes, $\epsilon=0$):

- **Final Success Rate:** ~60%

- **Consistency:** Evaluation curve is stable with no significant fluctuations

Part 3: Archer vs. Pigeon

Game Rule

Archer should shoot pigeon in 3 arrows.

Key Strategy

Archer Agent

ArcherAgent uses experience replay with parameter averaging:

- Records successful shots (adjustment, angle, predicted movement).
- On failure, converges toward average of successful cases.
- Stores data to dataset/successful_history (retains last 100 entries)

Pigeon Agent

PigeonAgent uses strategy-based learning:

- Evaluates danger zones using predicted aim line: $\text{archer_y} + dx \times \tan(\text{bow_angle})$.
- Learns safe Y-range (safe_y_min , safe_y_max) and danger_threshold via trial and error.
- Retains last 50 successful survival strategies.

Performance

1. **Archer Training (Stages 1 & 2.5)**
 - Pre-training success rate: ~10-20%
 - Post-training success rate: ~40-60%
 - Primarily learns **height_adjustment** to compensate for gravity
2. **Pigeon Training (Stage 2)**
 - Higher difficulty facing trained Archer
 - Expected success rate: ~30-50%
 - Learns to move within safe range and evade
3. **Final Test (Stage 3)**
 - Both use trained parameters
 - Win rate close to 50-50 (ideal balance)